

Andre Giske

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EDUCATION

University of California, San Diego

Sep. 2023 – June 2027

Bachelor of Science in Computer Science & Engineering

La Jolla, CA

GPA: 3.2

Relevant Coursework: Design & Analysis of Algorithms, Parallel Computing, Database System Principles, Large Codebases, Online Database Analytics & Applications, Operating Systems (OS), Computer Networks

TECHNICAL SKILLS

Languages: Java, Python, C/C++, Assembly, HTML, CSS, JavaScript, TypeScript, SQL

Frameworks & Libraries: FastAPI, Flask, Express, scikit-learn, SQLAlchemy, NumPy, Pydantic, JWT, bcrypt, React, Axios, Tailwind CSS, Socket.IO, Chart.js, SQLite

Developer Tools: Git, Docker, AWS, PostgreSQL, Nginx, Apache, Linux, VS Code, Kaggle

EXPERIENCE

Association for Computing Machinery (ACM)

May 2025 – Present

AI Competitions Team

San Diego, CA

- Reduced full tournament evaluation time from **~11 hours** to **~20 minutes** by designing a parallelized game orchestration system using **Python** multiprocessing, distributing **79,600** game evaluations across **32 worker processes** for a **32× speedup**.
- Architected and containerized a ML competition evaluation platform using **Docker**, **Flask** HTTP services, and a central game orchestrator, enabling language-agnostic bot submissions while maintaining complete game state isolation and eliminating turn-order race conditions to evaluate **200+** competitor submissions.

Mirron

Jan. 2026 – Mar. 2026

Startup Software Engineering Intern

Remote

- Architected and deployed **AWS cloud infrastructure** (S3, Control Tower, IAM) to configure **Electronic Lab Notebooks (ELNs)** for biologists studying gene activation via fluorescence imaging, enabling secure, scalable research workflows.
- Provisioned high-compute **AWS workstations** and configured organizational authorization policies in an 1 week **Agile environment**, expanding the team's computational capacity for data-intensive biological research.

PROJECTS

The Fray | *Python, FastAPI, PostgreSQL, scikit-learn, React, Axios*

May 2025 – Apr. 2026

Personalized News Recommendation App – www.thefraynews.com

- Improved article recommendation relevance by **~35%** by implementing a **KNN** collaborative filtering model with **scikit-learn**, training on user interaction vectors (clicks, read-time, saves) to surface personalized feeds across **50+ active users**.
- Modeled and managed a relational **PostgreSQL** schema with **SQLAlchemy** ORM, designing normalized tables for users, articles, and interaction history to support efficient personalized feed queries.
- Implemented token-based authentication including signup, login, and protected route middleware using **JWT** and **bcrypt** password hashing, eliminating server-side session state entirely.
- Deployed a production web application on **Nginx** with **SSL** termination, serving **100+ concurrent users** with zero-downtime configuration on a **Linux** VPS.

Sales Dashboard | *Node.js, Express, SQLite, Chart.js, Apache*

Jan. 2026 – Mar. 2026

Full-Stack Analytics Dashboard

- Structured a **Node.js** + **Express** web application using the **MVC** design pattern, cleanly separating routing, business logic, and data access layers to improve scalability and maintainability while enabling independent testing of each component.
- Configured an **Apache** HTTPS Server with structured access log pipelines, capturing per-request metadata to surface traffic patterns, peak usage windows, and session analytics using **Chart.js**.

Reliable Transport Protocol Simulator | *C programming, Docker*

Mar. 2026 – May 2026

Multi-host TCP Transport Layer

- Engineered a 3 state TCP congestion control state machine (Slow Start, AIMD, Fast Retransmit/Recovery) in **C programming**, tracking per-destination cwnd and ssthresh independently for up to **4 concurrent** hosts.